NETI PRELIMS REVISION SERIES

Topic 05 | Climatic Patterns of India

नेति नेति — Less noise. More clarity.

Why this matters for Prelims: Climate is one of the most Statement I/II-prone topics in the paper. Geography questions have risen from ~9 to ~13 per year and grown significantly harder since 2022. UPSC no longer asks "what is a monsoon" — it asks why a specific region has a rainfall anomaly, or whether a causal explanation between two climate statements is valid. Köppen classification, monsoon mechanics, and Western Disturbances each appear in almost every paper, often in the analytical format.

Monsoon onset sequence: Kerala (June 1) → Mumbai (June 10) → Delhi (June 29) → Rajasthan/J&K (July). Withdrawal is reverse — Rajasthan first (Sept), Kerala last (Oct end). UPSC has tested both onset and withdrawal sequences.

Köppen quick key: "A" = Tropical (no winter below 18°C) | "B" = Dry (arid/semi-arid) | "C" = Temperate | "D" = Continental | "E" = Polar/Tundra. India spans A, B, C, and E (Ladakh). No D climate in India.

Western Disturbances memory: Mediterranean origin → enter India from NW (Punjab/J&K) → winter rain in north India and snowfall in Himalayas → critical for rabi crops (wheat). Summer WDs = nor'westers (Kal Baisakhi) in eastern India — different origin.

THE SOUTHWEST MONSOON — MECHANICS & ANOMALIES

Aspect	Key Facts
Cause	Differential heating of land and sea. Indian subcontinent heats faster than Indian Ocean in summer → low pressure over land → moist winds from high-pressure ocean rush inland. The Himalayas prevent these moisture-laden winds from escaping northward.
Two branches	Arabian Sea branch: hits Kerala, moves up west coast and Western Ghats, crosses Vindhyas into north India. Bay of Bengal branch: enters via northeast India and Bangladesh, moves west. They meet over north-central India. Bay of Bengal branch arrives first.
Onset dates	Kerala: June 1 (±7 days). Goa: June 5–7. Mumbai: June 10. Delhi: June 29. Rajasthan and J&K: July. These are normal onset dates — year-to-year variation exists.
Withdrawal	Reverses the onset sequence. Begins from northwest (Rajasthan, ~Sept 1). Delhi: mid-Sept. Completes over Kerala by end of October. Total withdrawal ~45 days.
Rainfall distribution	Western Ghats windward (west): heavy — Cherrapunji/Mawsynram, Sahyadri slopes. Eastern Ghats leeward (east): rain-shadow — Deccan Plateau gets less. Northeast India: heaviest in India (Meghalaya). Rajasthan: least rainfall (<25 cm in Jaisalmer). Lakshadweep gets SW monsoon; Andaman & Nicobar also SW monsoon.
Orographic rainfall	Moist air forced upward by a mountain range → cools → condenses → precipitation on windward side. Leeward (rain shadow) side gets dry air. Western Ghats: windward = west

	(heavy rain), leeward = east (Deccan Plateau, dry). This is the mechanism UPSC uses in Statement I/II.
Break in monsoon	Temporary pause in monsoon activity when the trough (low-pressure axis) shifts north to the Himalayas. Foothills get rain; plains do not. Different from withdrawal — the monsoon has not retreated, just shifted.
El Niño effect	El Niño (warming of central/east Pacific) weakens the Indian monsoon — below-normal rainfall. La Niña (cooling) strengthens it — above-normal. ENSO (El Niño Southern Oscillation) is the broader phenomenon. 2023 El Niño caused deficient monsoon in parts of India — tested in 2024 paper context.

NOTE: Cherrapunji holds the record for most rainfall in a single month/year (historical records). Mawsynram holds the highest average annual rainfall record currently. UPSC has used both in wrong options.

NORTHEAST MONSOON, WESTERN DISTURBANCES & LOCAL WINDS

Phenomenon	What It Is / Why It Matters
Northeast Monsoon	Oct–Dec. Winds blow from land to sea (northeast → Bay of Bengal). Picks up moisture over Bay of Bengal → gives winter rain to Tamil Nadu and south Andhra. Chennai gets ~60% of its annual rainfall during this season — the single most tested monsoon anomaly fact. Rest of India has dry conditions.
Western Disturbances (WD)	Extratropical cyclones originating over the Mediterranean Sea (not Caspian or Atlantic — common UPSC trap). Travel eastward, enter India via northwest (Punjab, J&K, Himachal Pradesh). Bring winter rain (Dec–Feb) to north India plains and snowfall to Himalayas. Critical for rabi crops (wheat, mustard, barley). Weaker WDs = poor rabi season.
Kal Baisakhi (Nor'westers)	Pre-monsoon thunderstorms in West Bengal, Bihar, and Assam (April–May). Caused by strong convection over Chotanagpur plateau meeting moist Bay of Bengal air. Beneficial for tea (Assam) and jute. These are not Western Disturbances — different mechanism, different season, different region.
Mango Showers	Pre-monsoon showers in Kerala and Karnataka (April–May). Help mango ripening and coffee flowering. Local convective phenomenon, not related to WDs or NE monsoon.
Blossom Showers	Pre-monsoon showers in Kerala — important for tea and coffee plantation flowering. Often asked alongside Mango Showers as a matching pair.
Loo	Hot, dry, dust-laden wind blowing from west/northwest over the Indo-Gangetic Plain during May–June. Temperature can reach 45–48°C. Associated with heat wave conditions. Not a monsoon phenomenon — occurs before monsoon onset.
Mistral / Sirocco	Not Indian phenomena — but UPSC asks matching pairs of world winds. Mistral: cold north wind, France (Rhône Valley). Sirocco: hot, dry, dusty, North Africa → Mediterranean. Foehn: warm, dry, leeward side of Alps. Chinook: warm, dry, eastern Rockies (Canada/USA).

NOTE: UPSC distinguishes Kal Baisakhi (eastern India, pre-monsoon, convective) from Western Disturbances (northwest India, winter, Mediterranean origin). Confusing them is a recurring wrong-option trap.

KÖPPEN CLIMATE CLASSIFICATION — INDIA

Köppen uses vegetation as proxy for climate. India spans five major climate types. UPSC tests region → type matching and the criteria that distinguish types within the same group.

Code	Name	Key Criterion	Indian Regions	Distinguishing Feature
Af	Tropical Rainforest	All months >60 mm rain	Andaman & Nicobar Islands, Western Ghats (limited patches)	No dry season. Rainfall every month. Dense evergreen forest.
Am	Tropical Monsoon	Short dry season compensated by heavy monsoon	Malabar Coast, coastal Karnataka, most of Northeast India	Brief dry period (<2 months) but total rainfall so high that soil moisture is maintained. Distinguishable from Aw by rainfall totals.
Aw	Tropical Savanna	Dry winter season (>1 driest month <60 mm)	Most of peninsular India, interior Maharashtra, Odisha, parts of MP	Distinct wet and dry season. Deciduous vegetation. Most common Köppen type in India by area.
BSh	Hot Semi-Arid (Steppe)	Rainfall < 50% of potential evapotranspiration	Rajasthan (excluding Thar), parts of Gujarat, Deccan interior	Hot and dry. Not as extreme as BWh. Scrub vegetation. Rainfall 25–50 cm.
BWh	Hot Desert	Rainfall < 25% of potential evapotranspiration	Thar Desert (western Rajasthan, Jaisalmer area)	True hot desert. <25 cm annual rainfall. Very high summer temperatures.
Cwg	Monsoon-influenced humid subtropical	Warm summer, dry mild winter; "g" = Gangetic Plain	Indo-Gangetic Plain — Punjab, UP, Bihar, Bengal	"g" suffix is specific to Gangetic type: hottest month before the rainy season. Hot humid summers, cool dry winters.
Dfc	Subarctic / Boreal	Fewer than 4 months above 10°C	Higher Himalayas, parts of HP and Uttarakhand at high altitude	Short cool summers, very cold long winters. Snow for most of year.
ET	Tundra	Warmest month between 0°C and 10°C	Ladakh (cold desert, high altitude), Siachen area	Cold desert. No month above 10°C. Sparse vegetation. Ladakh = cold desert (low temp, very low rainfall).

NOTE: India has no "D" (continental) climate strictly speaking. The Himalayan/Ladakh zones use Dfc and ET. Thar = hot desert (BWh), Ladakh = cold desert (ET) — completely different mechanisms. UPSC has asked this distinction.

RAINFALL DISTRIBUTION — QUICK REFERENCE

Category	Region / Place	Exam-Ready Fact
Highest avg. annual rainfall	Mawsynram, Meghalaya	~11,872 mm/year. Located on southern face of Meghalaya hills — orographic effect from Bay of Bengal winds. Holds current highest annual rainfall record (not Cherrapunji).
Wettest single month/year (historical)	Cherrapunji, Meghalaya	Holds records for rainfall in a single month and single year. Both Mawsynram and Cherrapunji are in East Khasi Hills, Meghalaya.
Lowest rainfall	Jaisalmer (Rajasthan) / Rupshu (Ladakh)	Jaisalmer: ~9–12 cm/year (Thar Desert, SW monsoon weakens). Ladakh: even lower but cold desert. In peninsular India, Bellary (Karnataka) gets lowest.
NE monsoon rainfall	Tamil Nadu coast, SE Andhra	Chennai: ~60% rain from NE monsoon (Oct–Dec). Opposite of rest of India (SW monsoon = June–Sept). Reason: lies in rain shadow during SW monsoon; Bay of Bengal branch gives NE monsoon.
Rain shadow zone	Deccan Plateau (leeward of W. Ghats)	Western Ghats block SW monsoon moisture. Leeward (eastern) side — Pune receives less rain than Mumbai. Bengaluru (~970 mm) vs Mangaluru (~3500 mm).
Highest rainfall on Ghats	Agumbe (Karnataka) / Mahabaleshwar (MH)	Agumbe: "Cherrapunji of the South." King cobra habitat. Western Ghats windward face.
Winter rainfall	J&K, Punjab, HP, parts of Haryana/UP	From Western Disturbances only. Critical for wheat (rabi). TN also gets winter rain but from NE monsoon — different source.

STATEMENT I / II PRACTICE PAIRS — CLIMATE

These are the exact reasoning patterns UPSC applies. For each pair, identify: (a) is each statement true/false? (b) does II correctly explain I?

Statement I	Statement II	Answer & Why
Western Ghats receive heavy rainfall on the western slope.	The western slope faces the southwest monsoon winds and forces moist air upward, causing orographic precipitation.	Both TRUE. II correctly explains I. Answer: (a). Classic orographic rainfall question.
Tamil Nadu receives most of its rainfall during winter months.	Tamil Nadu is located on the windward side of the northeast monsoon.	Both TRUE. II correctly explains I. Answer: (a). NE monsoon picks up moisture over Bay of Bengal; TN coast faces it directly.

Ladakh is one of the coldest and driest places in India.	Ladakh lies in the rain shadow of both the Himalayas and the Karakoram Range.	Both TRUE. II correctly explains I. Answer: (a). Double rain shadow = minimal precipitation; high altitude = extreme cold.
Jaisalmer receives very low rainfall despite being in the path of the Arabian Sea monsoon branch.	The Arabian Sea monsoon branch loses most of its moisture after crossing the Aravalli Hills.	I is TRUE. II is PARTIALLY misleading — Aravallis run parallel (SW to NE), not perpendicular, so they do not effectively block moisture. The real reason is distance + Thar heat. Answer: (c) I correct, II incorrect.
A "break" in the monsoon does not mean the monsoon has withdrawn from India.	During a break, the ITCZ shifts toward the Himalayas, causing rainfall over foothills while the plains remain dry.	Both TRUE. II correctly explains I. Answer: (a). Break = trough shift north. Withdrawal = actual retreat southward.
Cherrapunji holds the record for the highest average annual rainfall in India.	Cherrapunji is located on the windward side of the Meghalaya hills facing Bay of Bengal winds.	I is FALSE (Mawsynram holds this record now). II is TRUE. Answer: (d) I incorrect, II correct. High-frequency UPSC trap.

NOTE: Always check Statement I first, independently. Never assume it is true because Statement II sounds like a good explanation. UPSC regularly makes I false and II true — option (d).

RECENT IN NEWS (2023–2025)

Event / Issue	Climate Angle — Why UPSC-Relevant
El Niño 2023–24	Strong El Niño weakened SW monsoon in 2023 (below-normal year). IMD issued forecasts citing ENSO. Tests: ENSO mechanism, El Niño vs La Niña effect on Indian monsoon, Indian Ocean Dipole (IOD) as countervailing factor.
Indian Ocean Dipole (IOD)	Positive IOD (western Indian Ocean warmer than east) enhances Indian monsoon. Negative IOD weakens it. 2019 had extreme positive IOD — buffered El Niño impact. IOD increasingly tested in Statement I/II format with ENSO.
Heatwave records 2024	2024 saw earliest and most intense heatwaves in parts of north India. Tests: IMD definition of heatwave (temperature ≥40°C in plains and ≥5°C above normal). Loo winds as mechanism. NDMA guidelines on heatwave — disaster management angle.
Cyclone activity 2023–25	Cyclone Biparjoy (2023, Arabian Sea), Cyclone Michaung (2023, Bay of Bengal). Tests: Arabian Sea cyclones are rarer but more intense. NE coast (AP, Odisha) most cyclone-prone. IMD naming system. Pre-monsoon vs post-monsoon cyclone seasons.
Cloud seeding — Maharashtra	Maharashtra undertook cloud seeding (silver iodide) during 2023 drought. Tests: mechanism of artificial rainfall, silver iodide as nucleating agent, limitations (needs existing cloud cover). Statement I/II potential: "Cloud seeding creates rain from clear sky" (FALSE).

IMD centenary (2025)	India Meteorological Department celebrated its 150th year in 2025. IMD under MoES (Ministry of Earth Sciences). Responsible for monsoon, cyclone, earthquake, and tsunami monitoring. Institutional knowledge question potential.

★ PYQ Insight:

2024 — Statement I/II: "Statement I: Some regions receive less rainfall in the presence of cyclones. Statement II: Cyclones draw moisture away from surrounding areas." Both true — but II does NOT correctly explain I. Correct answer: (b) Both correct, II does not explain I. Classic post-2020 trap — students who only check truth of each statement get it wrong.

2023 — Matching pairs of Köppen climate types with Indian regions. Tested whether students knew Aw (tropical savanna) vs Am (tropical monsoon) distinction, and Bsh (hot semi-arid) for Rajasthan. Broad labels alone are insufficient.

2022 — Statement-based on Western Disturbances: "Western Disturbances originate over the Caspian Sea" (FALSE — Mediterranean Sea/Atlantic). "They bring snowfall to the Himalayas" (TRUE). UPSC exploits the Caspian vs Mediterranean confusion repeatedly.

2020 — Statement I/II (inaugural pivot year): "The Western Ghats receive more rainfall on the windward side. This is because the windward side forces air upward, causing orographic rainfall." Both true, II correctly explains I — answer (a). This is the prototype of how UPSC frames monsoon + orographic questions.

Recurring trap — Chennai receives most of its rainfall during the northeast monsoon (Oct–Dec), not the southwest monsoon like the rest of India. Also: Mawsynram/Cherrapunji distinction — Mawsynram (not Cherrapunji) currently holds the highest annual rainfall record. Both have appeared as wrong options.

3 THINGS UPSC TESTS FROM THIS TOPIC

- **Causal reasoning about rainfall anomalies.** UPSC does not ask "which region gets most rainfall." It asks WHY Chennai gets rain in winter, WHY the Deccan Plateau is dry, WHY Ladakh is a cold desert. For each rainfall anomaly in India, know the mechanism: orographic effect, monsoon branch, rain shadow, ENSO influence. Every anomaly is a potential Statement I/II pair. Prepare by asking "why" for each fact, not just "what."
- **Köppen type → region matching at the sub-type level.** Knowing broad groups (A = tropical, B = dry) is not enough. UPSC has tested Aw vs Am (savanna vs monsoon), BSh vs BWh (semi-arid vs desert), and the Cwg suffix specific to the Gangetic Plain. For India, you need: Af (A&N Islands), Am (Malabar/NE India), Aw (most peninsular India), BSh/BWh (Rajasthan), Cwg (IGP), ET (Ladakh). Fix these six and ignore the rest.
- **Western Disturbances: origin, season, region, and crop link.** Mediterranean origin (not Caspian — UPSC trap). Northwest India. December–February. Winter rain on plains, snowfall in Himalayas. Rabi crop connection (wheat). Know what happens when WDs are weak: poor rabi harvest, drought in north India. Also distinguish WDs from Kal Baisakhi (eastern India, pre-monsoon, convective — not Mediterranean). These appear together in matching pairs.

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